

The Temporal Tug-of-War: Detecting RAG Conflicts in Diffusion Language Models via Trajectory Variance

Anonymous ACL submission

Abstract

Retrieval-Augmented Generation (RAG) significantly improves the factual grounding of language models but introduces the risk of knowledge friction when retrieved contexts conflict with the model’s parametric memory. While autoregressive models resolve these conflicts causally, Discrete Diffusion Language Models (DLMs) refine sequences globally across a temporal denoising schedule, obfuscating the arbitration process. Inspired by recent trajectory-probing frameworks, we propose the Trajectory Variance Score (TVS), a novel methodology to detect parametric versus external memory conflicts. By modeling the diffusion process as a Markov Decision Process (MDP) and tracking multi-seed variance across the $T = 50$ stochastic denoising steps, we extract the temporal dynamics of model confidence. We feed these sequential variance features, along with their temporal derivatives, into an Attention-Augmented Bidirectional Long Short-Term Memory (LSTM) network for binary classification. Our empirical results on LLaDA and DREAM across four datasets (Synthetic, SciQ, PopQA, and CounterFact) demonstrate that TVS successfully captures the temporal "tug-of-war" of knowledge friction, achieving an AUROC of 0.79 on LLaDA and establishing a robust interpretability framework for diffusion-based text generation.

1 Introduction

The adoption of Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) has become the standard mechanism for mitigating hallucinations in large language models. However, this architecture introduces a secondary failure mode: knowledge conflict. When an injected RAG context directly contradicts the model’s pre-trained parametric memory (e.g., outdated facts or adversarially generated fakes), the model must implicitly arbitrate between its internal prior and the external evidence.

In standard autoregressive (AR) models, this friction is instantaneous and localized, manifesting as spikes in token-level perplexity during left-to-right decoding. However, the emergence of Discrete Diffusion Language Models (DLMs) like LLaDA (Nie et al., 2025) and DREAM (Ye et al., 2025) fundamentally alters this paradigm. Unlike AR models, DLMs generate text through an iterative, non-causal denoising process. The resolution of knowledge conflicts is therefore not a single token-emission event, but a temporal "tug-of-war" distributed across the entire diffusion trajectory.

Recent work in DLM interpretability, such as TraceDet (Chang et al., 2025), has demonstrated that hidden trajectory features can be extracted post-generation to classify intrinsic hallucinations. Building upon this trajectory-centric view, we introduce the **Trajectory Variance Score (TVS)**. We hypothesize that forced arbitration between contradictory memory sources induces measurable instability in the intermediate diffusion states. By executing multiple generation runs with varying seeds and extracting the variance of prediction confidences across $T = 50$ timesteps, we map the temporal topology of this uncertainty. We then utilize an Attention-Augmented Bidirectional LSTM network, processing both variance and velocity features, to classify these temporal signatures.

Our contributions are threefold: 1. We formally define the problem of RAG-induced knowledge friction within the discrete diffusion paradigm. 2. We propose TVS, a multi-seed trajectory probing technique that models the temporal arbitration of conflicting knowledge. 3. We provide extensive empirical evaluations on LLaDA and DREAM, demonstrating strong conflict detection capabilities across diverse datasets.

2 Related Work

2.1 Hallucination and Conflict Detection in AR-LLMs

Hallucination detection in autoregressive models has been extensively studied. Output-based detection leverages signals derived from the generated text, such as semantic entropy (Kuhn et al., 2023) or lexical similarity. The core intuition is that hallucinated or conflicting responses are associated with lower predictive confidence. Latent-based methods instead probe the hidden representations during a single forward pass, utilizing techniques like Contrast-Consistent Search (CCS) (Burns et al., 2023) to separate truthful from hallucinated states. However, these methods are uniquely suited to the left-to-right generation paradigm and fail to capture the multi-step refinement of diffusion models.

2.2 Discrete Diffusion Language Models

Discrete Diffusion Language Models have emerged as powerful alternatives to AR models (Sahoo et al., 2024; Shi et al., 2024). Models such as LLaDA (Nie et al., 2025) utilize discrete remasking processes to scale up to 8B parameters, achieving performance comparable to leading LLMs. Similarly, DREAM (Ye et al., 2025) adapts diffusion paradigms for enhanced reasoning. Recent efforts like SPREAD (Yu et al., 2026) have begun exploring DLMs within the RAG framework, identifying Response Semantic Drift (RSD) and employing query-relevance-guided denoising to keep generation anchored to the query semantics. However, while SPREAD focuses on actively mitigating drift during generation to improve precision, the specific mechanisms of detecting the underlying parametric versus external knowledge conflicts remain underexplored.

2.3 Trajectory Analysis in Diffusion Models

The shift to iterative refinement necessitates new interpretability tools. The most relevant precursor to our work is the TraceDet framework (Chang et al., 2025), which formulates the denoising process as an action trace and applies the Information Bottleneck principle to extract sub-traces informative of hallucinated outputs. Other contemporary methods have also focused on trajectory features: TDGNet (Hemmat et al., 2026) formulates hallucination detection by learning over evolving token-level attention graphs, while OSCAR (Shah et al., 2026) computes cross-chain Shannon

entropy across parallel denoising chains with randomized reveal orders to detect high-uncertainty positions. While TraceDet, TDGNet, and OSCAR effectively localize intrinsic hallucinations or general uncertainty, they do not isolate the specific phenomenon of RAG-induced knowledge friction. Our work extends trajectory analysis by specifically modeling the temporal dynamics of multi-seed variance—capturing both raw variance and its velocity across the full schedule—to detect the unique "tug-of-war" signature of parametric conflicts.

3 Background: Diffusion Large Language Models

3.1 Formal Formulation of D-LLMs

Unlike AR models that rely on the next-token prediction paradigm $P(x_i|x_{<i})$, DLMs generate responses by iterative refinement through a forward noising and backward denoising process over T time steps.

Let $r = (r_0, \dots, r_T)$ denote the sequence of intermediate texts, where each $r_t \in \mathcal{V}^n$ is a token sequence of length n . Here, r_0 represents the fully clean, structured text sequence, and r_T represents the fully masked sequence (pure noise).

Given an input query p_0 , the **forward noising process** is defined by a sequence of transition distributions $\{q(r_t|r_{t-1})\}_{t=1}^T$, mathematically expressed as:

$$q(r_{1:T}|r_0) = \prod_{t=1}^T q(r_t|r_{t-1}) \quad (1)$$

This process gradually corrupts the sequence by replacing tokens with a designated '[MASK]' state.

Conversely, the **reverse denoising process** is parameterized by a neural network defining a sequence of conditional distributions $\{P_\theta(r_0|r_t, p_0)\}_{t=1}^T$. At each timestep $t > 0$, the model predicts all masked tokens simultaneously from the current state r_t , yielding an estimated clean sequence $\tilde{r}_{t-1} \sim P_\theta(r_0|r_t, p_0)$. Following this prediction, a discrete remasking strategy is applied—typically low-confidence remasking—where a specific fraction ρ_t of the predicted tokens are remasked to form the intermediate state r_{t-1} .

3.2 Denoising as a Markov Decision Process (MDP)

To understand how knowledge conflicts manifest in DLMs, it is advantageous to formulate the denoising process as a Markov Decision Process over

the decoding steps, expanding on the definitions utilized in recent trajectory analysis (Chang et al., 2025):

- **State:** The t -th state $s_t = (p_0, r_{T-t})$ in the denoising schedule is defined as the combination of the RAG context p_0 and the intermediate, partially masked sequence r_{T-t} .
- **Action:** At state s_t , the DLM takes an action a_t by predicting the identity of all masked tokens, generating a full token sequence proposal $\hat{r}_{T-t-1}$ from the parameterized distribution $P_\theta(r_0|r_{T-t}, p_0)$.
- **Transition:** Following the action, a fraction of the tokens in $\hat{r}_{T-t-1}$ are remasked based on the predefined noise schedule ρ_t . The DLM then transitions to the next state $s_{t+1} = (p_0, r_{T-t-1})$.

Under this MDP framework, a single generation run yields a complete *action trace* $A = \{a_0, a_1, \dots, a_{T-1}\}$.

3.3 Knowledge Friction in the Action Trace

We define *Knowledge Friction* as the latent instability generated during the action a_t when the input condition p_0 (the retrieved context) contains a proposition that directly contradicts the model’s learned parametric weights θ_{true} .

For instance, consider the query: “What is the capital of France?” The model’s parametric memory strongly associates this with “Paris” (θ_{true}). However, if a conflicting RAG context states, “The capital of France is London” (p_0), a direct contradiction occurs.

In standard generations (where p_0 aligns with θ_{true}), the sequence of actions A converges smoothly, as the gradients from the context and the parametric memory pull the sequence toward the same semantic attractor (e.g., “Paris”). However, under a RAG conflict, the model experiences a “tug-of-war.” The transition dynamics are disrupted because the action a_t oscillates between satisfying the external context p_0 (e.g., “London”) and the internal prior θ_{true} (e.g., “Paris”). Because DLMs utilize parallel decoding, this friction is not isolated to a single token but is distributed globally across the intermediate action trace.

4 Methodology: Trajectory Variance Score

To detect this knowledge friction, we must isolate the temporal instability it creates. We achieve this by analyzing the variance in model confidence across multiple stochastic denoising runs, effectively measuring the divergence of parallel action traces. The end-to-end pipeline of our framework, from dataset filtering to final classification, is illustrated in Figure 1.

End-to-End Evaluation Pipeline for Knowledge Friction

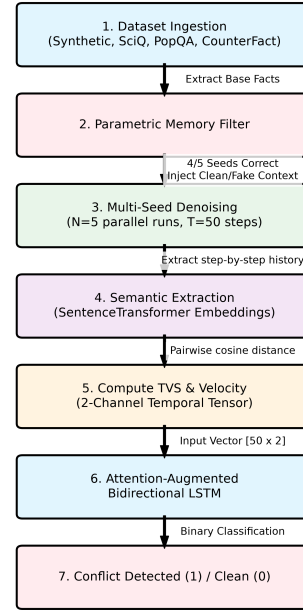


Figure 1: End-to-end evaluation pipeline: Data points undergo strict parametric memory filtering before being injected with RAG fakes. We extract 5 parallel trajectories, compute raw variance and velocity features, and classify them using an Attention-Augmented Bidirectional LSTM.

4.1 Parametric Memory Filtering and Trajectory Generation

Diffusion models are inherently stochastic; the specific trajectory an action trace A takes from s_0 (pure noise) to s_T (final text) depends heavily on the initial random seed governing the remasking logic. Before generating conflict trajectories, we establish a rigorous baseline by filtering the dataset. A fact is only retained if the base DLM correctly generates the true answer from its parametric memory across at least 4 out of $N = 5$ parallel seeds. This ensures any subsequent friction is genuinely induced by the RAG context rather than inherent model ignorance.

For a verified RAG prompt p_0 , we execute $N = 5$ independent denoising runs over $T = 50$ diffusion timesteps. For each independent run i and timestep t , we extract the model’s semantic prediction for the target entity and embed it to compute the confidence $P_{i,t}$.

4.2 TVS Formulation and Velocity

In the absence of knowledge friction, all N runs rapidly converge on identical semantic structures, yielding low variance across the parallel traces. Under conflict, the competing pulls of the parametric prior and the injected context cause the unmasking trajectories to diverge significantly before finally crystallizing.

We define the Trajectory Variance Score (TVS) at timestep t as the unbiased sample variance of the semantic predictions across the N independent runs:

$$\text{TVS}_t = \frac{1}{N-1} \sum_{i=1}^N (P_{i,t} - \bar{P}_t)^2 \quad (2)$$

where $\bar{P}_t$ is the mean semantic prediction at step t .

Crucially, the raw variance magnitude only captures a static snapshot of uncertainty. The *rate of change* in variance is equally critical for detecting sudden semantic shifts during the "tug-of-war." Thus, we compute the temporal derivative (velocity) of the variance:

$$\Delta \text{TVS}_t = \text{TVS}_t - \text{TVS}_{t-1} \quad (3)$$

Computing these across all timesteps yields a 2-channel temporal feature vector $\mathbf{V} = [(\text{TVS}_1, \Delta \text{TVS}_1), \dots, (\text{TVS}_{50}, \Delta \text{TVS}_{50})] \in \mathbb{R}^{T \times 2}$, which maps the topological landscape and dynamics of generation uncertainty.

4.3 Attention-Augmented Classification Framework

Because the resolution of a RAG conflict is a sequential process—exhibiting phases of early exploration, peak friction during the mid-steps, and final crystallization—we formulate conflict detection as a sequence classification task using an **Attention-Augmented Bidirectional LSTM**.

The 2-channel trajectory vector $\mathbf{V}$ is passed through a 2-layer Bidirectional LSTM (hidden size = 32 per direction), allowing the model to contextualize variance spikes globally across the denoising schedule. To explicitly capture the critical "mid-step" friction phases, the recurrent outputs are

processed by a dynamic sequence attention layer, producing a weighted context vector. This representation is then fed into a Multi-Layer Perceptron (MLP) head with batch normalization and dropout to prevent overfitting, outputting a binary classification $\hat{y} \in \{0, 1\}$, where 1 indicates a RAG conflict.

5 Experimental Setup

5.1 Models and Datasets

We evaluate TVS on two recent DLMS: **LLaDA** and **DREAM**. To establish a robust testbed, we utilize four diverse datasets: **Synthetic** (custom factual queries), **SciQ** (Welbl et al., 2017), **PopQA** (Mallick et al., 2024), and **CounterFact** (Meng et al., 2022).

For each dataset, we first filter for baseline instances where the base model accurately predicts the True Answer from its parametric memory with high confidence. We then synthetically inject a contradictory RAG Fake into the context window. The final test dataset is strictly balanced between Clean ($y = 0$) and Conflict ($y = 1$) samples.

5.2 Implementation Details

Generations were executed using $N = 5$ seeds over a discrete diffusion schedule of $T = 50$ timesteps. The Attention-Augmented Bidirectional LSTM classifier comprises 2 hidden layers with 32 units per direction. It was trained using Binary Cross-Entropy loss, optimized via AdamW with a weight decay of $1e-4$, and employed a Cosine Annealing learning rate schedule over 75 epochs. To ensure robust evaluation, we utilized a strict 70/15/15 stratified split for the training, validation, and testing sets.

6 Results

6.1 Overall Detection Performance

The aggregate performance of the TVS-LSTM detector on the test set is detailed in Table 1. The framework achieves strong performance on both models. LLaDA demonstrates higher overall detectability (71.19% Accuracy, 0.7875 AUROC) compared to DREAM (70.18% Accuracy, 0.7679 AUROC). Notably, the precision for detecting conflicts ($y = 1$) is consistent across both models (0.71 for LLaDA, 0.71 for DREAM), indicating that when TVS identifies a variance spike in the action trace, it is a reliable indicator of parametric friction.

Model	Acc (%)	AUROC	Prec	Rec	F1
LLaDA	71.19	0.7875	0.71	0.71	0.71
DREAM	70.18	0.7679	0.71	0.68	0.70

Table 1: Overall conflict detection performance on the test set. Precision, Recall, and F1 are reported for the Conflict (1) class.

6.2 Dataset-Specific Dynamics

Table 2 provides the accuracy breakdown across the four evaluated datasets.

Model	Synth.	SciQ	PopQA	CounterFact
LLaDA	79.67	75.41	71.51	63.62
DREAM	82.76	70.57	69.52	59.24

Table 2: Detection accuracy (%) breakdown across specific dataset domains.

The results reveal model-specific sensitivities to the domain of knowledge. While both models perform exceptionally well on the Synthetic dataset (approx. 80% - 83%), both struggle with the CounterFact dataset, particularly DREAM (59.24% accuracy). This suggests that for DREAM, counterfactual injections either override the parametric memory without triggering significant friction, or the conflict is resolved too early in the MDP trace to be fully captured by standard variance thresholding. Conversely, LLaDA maintains slightly better detectability (63.62%) on counterfactuals, while showing relatively consistent performance (71% - 80%) across other domain types.

7 Analysis: Interpreting the Tug-of-War

Visualizing the average TVS curves in Figure 2 provides crucial interpretability into the MDP trajectory, revealing that knowledge friction manifests as a sustained elevation in generation uncertainty.

As shown in Figure 2, in non-conflict scenarios (blue solid lines), the models follow a standard denoising trajectory where semantic variance decays as the generation crystallizes. Under a RAG conflict (red dashed lines), the TVS trajectory exhibits a distinct, sustained gap of elevated variance.

For LLaDA, the competing gradients of the parametric prior and the RAG fake clash extensively during the early-to-mid steps ($t \in [0, 25]$). This conflict maintains a significantly higher cross-seed variance and delays the final crystallization. Even after the model mostly converges ($t > 30$), a resid-

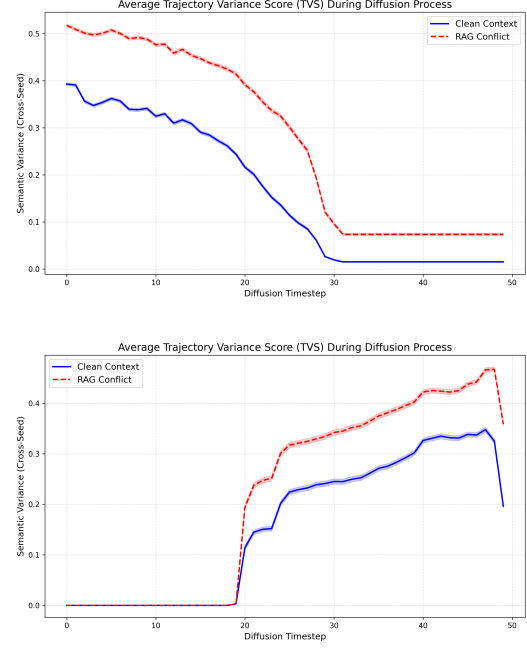


Figure 2: Average TVS trajectories across denoising steps for LLaDA (top) and DREAM (bottom). The sustained elevation in cross-seed variance is clearly visible under RAG conflict compared to the clean baseline.

ual variance gap persists. DREAM exhibits a remarkably similar separation during its active semantic resolution phase ($t > 20$, noting the inverted generation step indexing). In both models, it is this persistent "tug-of-war"—characterized by elevated variance magnitude and delayed convergence—that the Attention-Augmented LSTM successfully captures as the temporal signature of parametric friction.

8 Conclusion

This paper introduces the Trajectory Variance Score (TVS), an interpretability framework for detecting RAG conflicts in Discrete Diffusion Language Models. By formalizing generation as an MDP and capturing both the raw variance and velocity of multi-seed trajectories, we achieve robust detection of knowledge friction. Our implementation of an Attention-Augmented Bidirectional LSTM successfully models these temporal dynamics, establishing a powerful new probing technique for safeguarding the integrity of diffusion-based text generation.

References

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. 2023. [Discovering latent knowledge in lan-](#)

397	guage models without supervision . In <i>International</i>	449
398	<i>Conference on Learning Representations (ICLR)</i> .	450
399	Shenxu Chang, Junchi Yu, Weixing Wang, Yongqiang	451
400	Chen, Jialin Yu, Philip Torr, and Jindong Gu. 2025.	452
401	Tracedet: Hallucination detection in discrete diffu-	
402	sion models via hidden trajectories. <i>arXiv preprint</i>	453
403	<i>arXiv:2510.01274</i> .	454
404	Arshia Hemmat and 1 others. 2026. Tdgnet: Hal-	455
405	lucination detection in diffusion language mod-	456
406	els via temporal dynamic graphs. <i>arXiv preprint</i>	457
407	<i>arXiv:2602.08048</i> .	
408	Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023.	
409	Semantic uncertainty: Linguistic invariances for un-	
410	certainty estimation in natural language generation .	
411	In <i>The Eleventh International Conference on Learn-</i>	
412	<i>ing Representations</i> .	
413	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio	
414	Petroni, Vladimir Karpukhin, Naman Goyal, Hein-	
415	rich Küttler, Mike Lewis, Wen-tau Yih, Tim Rock-	
416	täschel, and 1 others. 2020. Retrieval-augmented	
417	generation for knowledge-intensive nlp tasks. <i>Ad-</i>	
418	<i>vances in Neural Information Processing Systems</i> ,	
419	33:9459–9474.	
420	Chaitanya Mallick and 1 others. 2024. Generate, re-	
421	trieve, or both? context-aware knowledge retrieval	
422	for open-domain question answering. <i>arXiv preprint</i>	
423	<i>arXiv:2411.16133</i> .	
424	Kevin Meng, David Bau, Alex Andonian, and Yonatan	
425	Belinkov. 2022. Locating and editing factual asso-	
426	ciations in GPT . <i>Advances in Neural Information</i>	
427	<i>Processing Systems</i> , 35:17359–17372.	
428	Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang,	
429	Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong	
430	Wen, and Chongxuan Li. 2025. Large language dif-	
431	fusion models. <i>arXiv preprint arXiv:2502.09992</i> .	
432	Subham Sekhar Sahoo, M. Arriola, Y. Schiff,	
433	A. Gokaslan, E. Marroquin, J. Chiu, A. Rush, and	
434	V. Kuleshov. 2024. Simple and effective masked	
435	diffusion language models . In <i>Advances in Neural</i>	
436	<i>Information Processing Systems</i> .	
437	Yash Shah, Abhijit Chakraborty, Naresh Kumar Devu-	
438	lapally, Vishnu Lokhande, and Vivek Gupta. 2026.	
439	Oscar: Orchestrated self-verification and cross-path	
440	refinement. <i>arXiv preprint arXiv:2604.01624</i> .	
441	Jiaxin Shi, Kehang Han, Zhe Wang, Arnaud Doucet, and	
442	Michalis K. Titsias. 2024. Simplified and generalized	
443	masked diffusion for discrete data . In <i>Advances in</i>	
444	<i>Neural Information Processing Systems</i> .	
445	Johannes Welbl, Nelson F Liu, and Matt Gardner. 2017.	
446	Crowdsourcing multiple choice science questions .	
447	In <i>Proceedings of the 3rd Workshop on Noisy User-</i>	
448	<i>generated Text</i> , pages 94–106.	
	Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui	
	Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong.	
	2025. Dream 7b: Diffusion large language models.	
	<i>arXiv preprint arXiv:2508.15487</i> .	
	Chuanyue Yu, Jiahui Wang, Yuhan Li, Heng Chang,	
	Ge Lan, Qingyun Sun, Jia Li, Jianxin Li, and Ziwei	
	Zhang. 2026. Unlocking the potentials of retrieval-	
	augmented generation for diffusion language models.	
	<i>arXiv preprint arXiv:2601.11342</i> .	